**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

**DEPARTMENT INDUSTRIAL VERTICAL - SYSTEM ON CHIP**

**LOW POWER VLSI DESIGN(BE23EC512)**

# ENERGY RECOVERY CIRCUIT WITH IMPROVED DIODE FREE ADIABATIC LOGIC


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**ENERGY RECOVERY CIRCUIT WITH IMPROVED DIODE FREE ADIABATIC LOGIC**

## 1)Inverter Energy Recovery Circuit Using Adiabatic Logic:

An energy recovery circuit with improved diode-free adiabatic logic (DFAL) eliminates diodes from the charging/discharging paths, using MOS transistors and a split-level power clock for energy-efficient switching. This approach minimizes power loss by avoiding diode threshold voltage drops and enables significant charge recovery during logic transitions. The result is enhanced energy savings and lower dissipation compared to traditional diode-based adiabatic circuits.

### Circuit Diagram

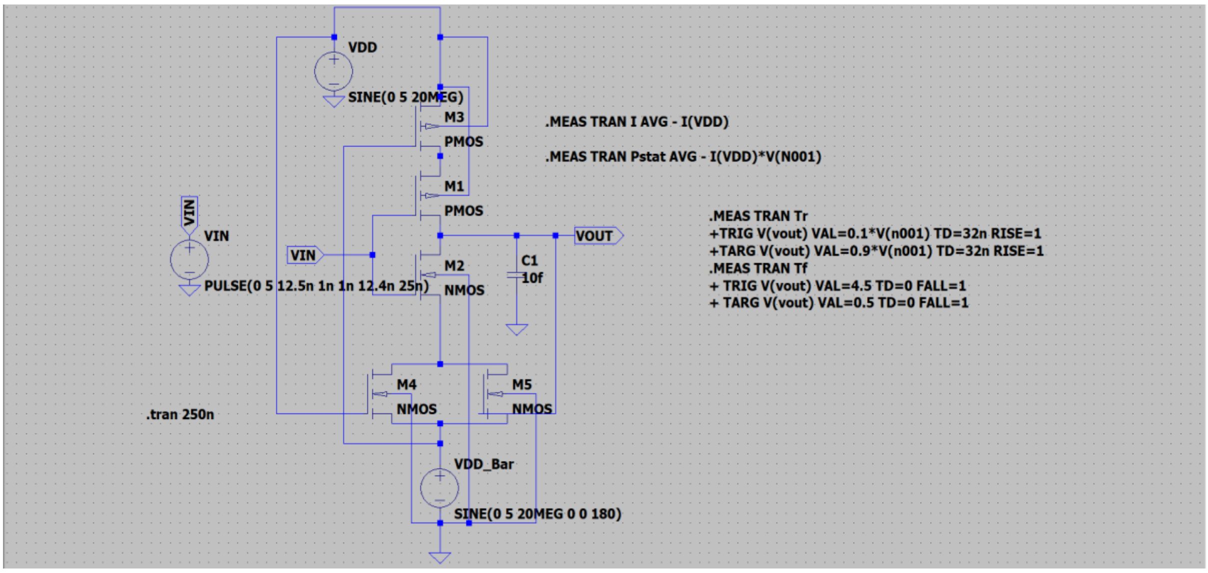


Figure 1 – Circuit Diagram of Energy Recovery Circuit

### Waveform

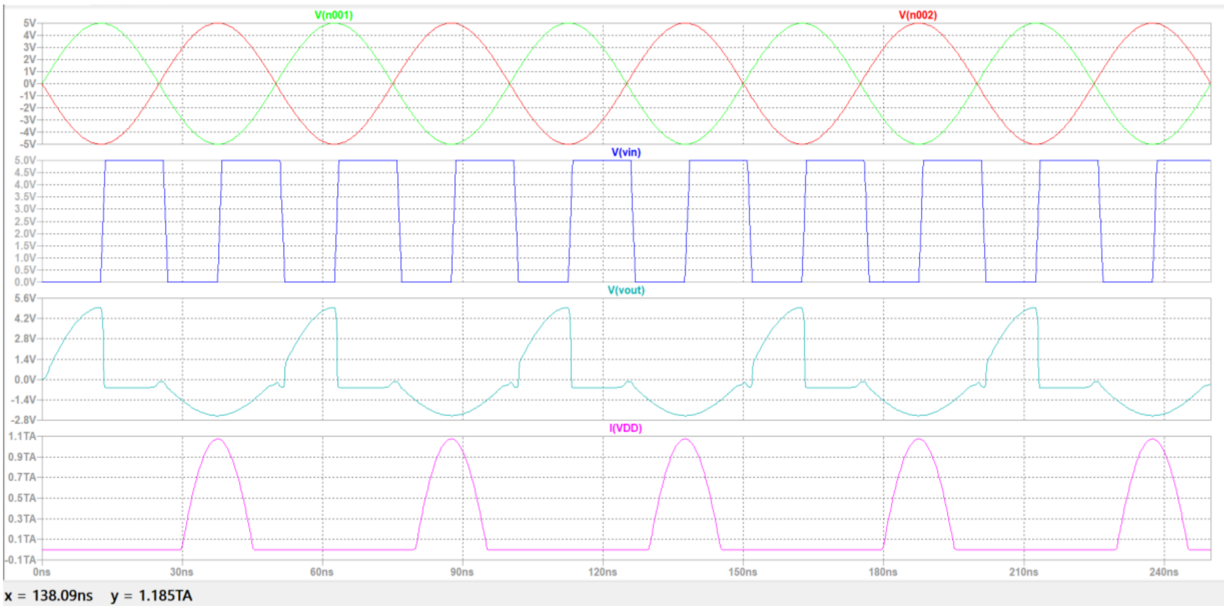


Figure 2 - Waveform of Energy recovery circuit

### 2)Analysing Transistor Sizing and Optimization (Symmetric in Nature):

It is a crucial for minimizing energy dissipation while ensuring correct logical functionality. The gate capacitance, which depends on transistor size, must be carefully balanced to optimize both speed and energy recovery—smaller transistors reduce capacitance and energy loss but may increase circuit delay. Designers must size transistors to achieve minimal power dissipation, efficient charge transfer, and acceptable timing performance in adiabatic circuit operation.

| Size | Pmos | Nmos |
|---|---|---|
| Width | 2u | 1u |
| Length | 0.18u | 0.18u |

### 3)Rise Time and Fall Time:

In improved diode-free adiabatic logic (IDFAL), rise time is the interval for a signal to transition from 10% to 90% of its peak value, while fall time is the reverse—going from 90% to 10%. These times directly impact the circuit's speed and energy recovery efficiency, since longer rise or fall times can improve charge recycling but may slow the operation. Careful control of these times is essential in IDFAL to balance low energy dissipation and desired speed.

**Rise Time (Manual)**

Tr=3.9933884e^10

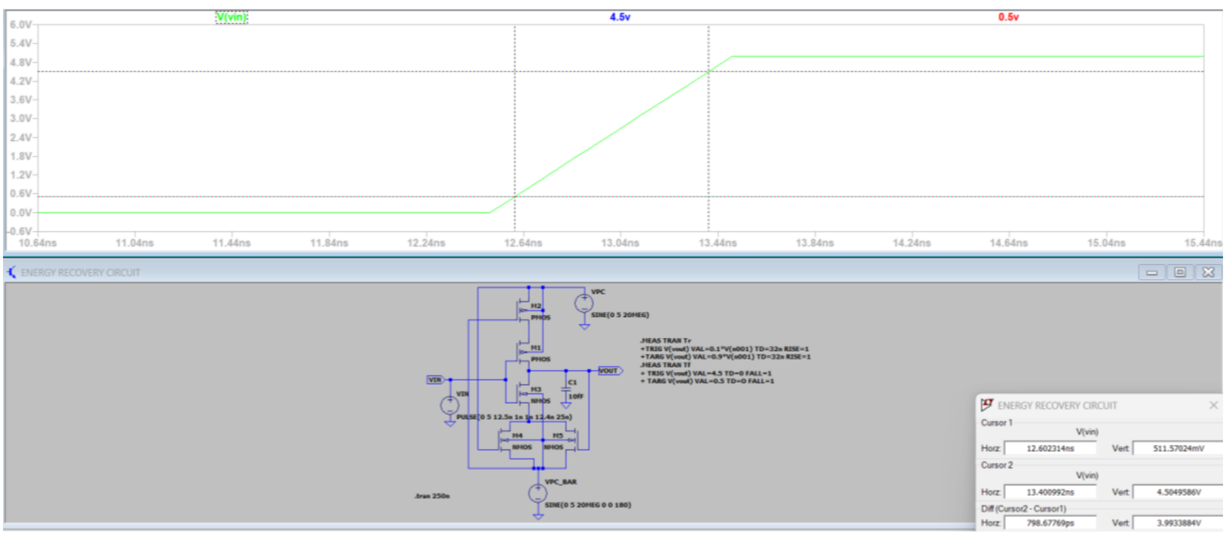


Figure 3 - Rise Time of Energy Recovery Circuit

### Fall Time (Manual)

Tf=3.991204e^10

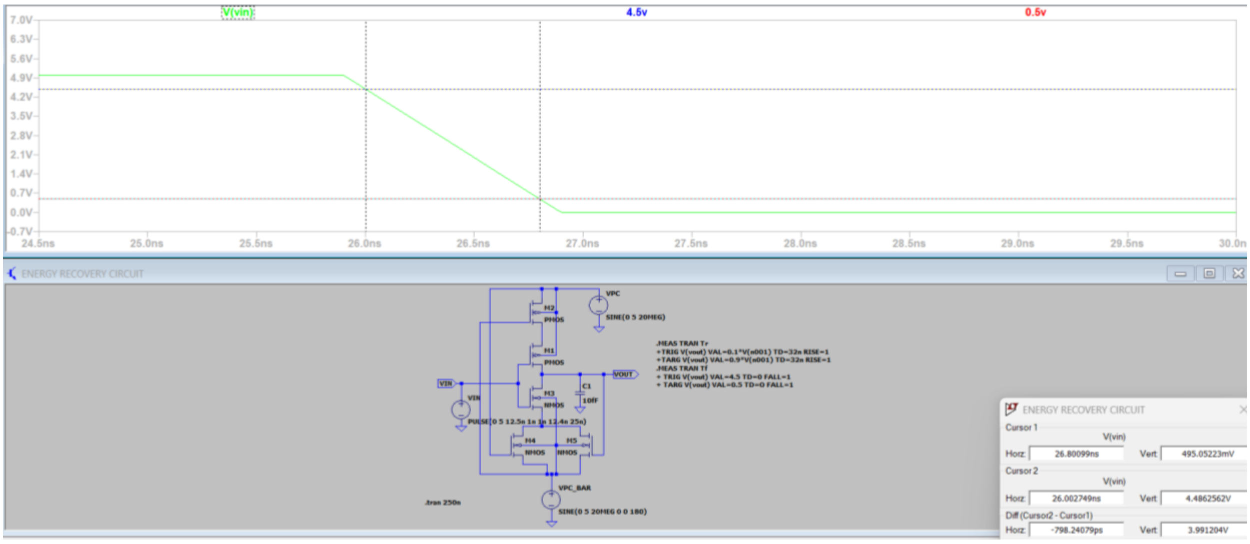


Figure 4 - Fall time of Energy Recovery Circuit

### Rise Time and Fall time Using Command Lines

Tr=3.38298437242 e^(-10)

Tf=3.36171611875 e^(-10)

```
SPICE Output Log: C:\Users\ASUS\OneDrive\Documents\LTspice\energy recovery.log
LTspice 24.1.9 for Windows
Circuit: C:\Users\ASUS\OneDrive\Documents\LTspice\energy recovery.net
Start Time: Wed Sep 24 15:14:24 2025
solver = Normal
Maximum thread count: 16
tnom = 27
temp = 27
method = trap
Instance "M2": Length shorter than recommended for a level 1 MOSFET.
Instance "M2": Width narrower than recommended for a level 1 MOSFET.
Instance "M4": Length shorter than recommended for a level 1 MOSFET.
Instance "M4": Width narrower than recommended for a level 1 MOSFET.
Instance "M5": Length shorter than recommended for a level 1 MOSFET.
Instance "M5": Width narrower than recommended for a level 1 MOSFET.
Instance "M1": Length shorter than recommended for a level 1 MOSFET.
Instance "M1": Width narrower than recommended for a level 1 MOSFET.
Instance "M3": Length shorter than recommended for a level 1 MOSFET.
Instance "M3": Width narrower than recommended for a level 1 MOSFET.
Direct Newton iteration for .op point succeeded.
Total elapsed time: 0.112 seconds.

Files loaded:
C:\Users\ASUS\OneDrive\Documents\LTspice\energy recovery.net
C:\Users\ASUS\AppData\Local\LTspice\lib\cmp\standard.mos

i: AVG(- I(VDD))=-215557156296 FROM 0 TO 2.5e-07
pstat: AVG(- I(VDD)*V(N001))=982648996382 FROM 0 TO 2.5e-07
tr=3.38298437242e-10 FROM 5.19078324126e-08 TO 5.22461308498e-08
tf=3.36171611875e-10 FROM 1.28170619054e-08 TO 1.31532335173e-08
```

## 4)Noise Margin:

It is defined as the difference between the actual output voltage and the minimum input voltage required to recognize a logic high or low correctly. Larger noise margins indicate better immunity to noise, ensuring reliable circuit operation despite voltage fluctuations or disturbances.

### Circuit and waveform

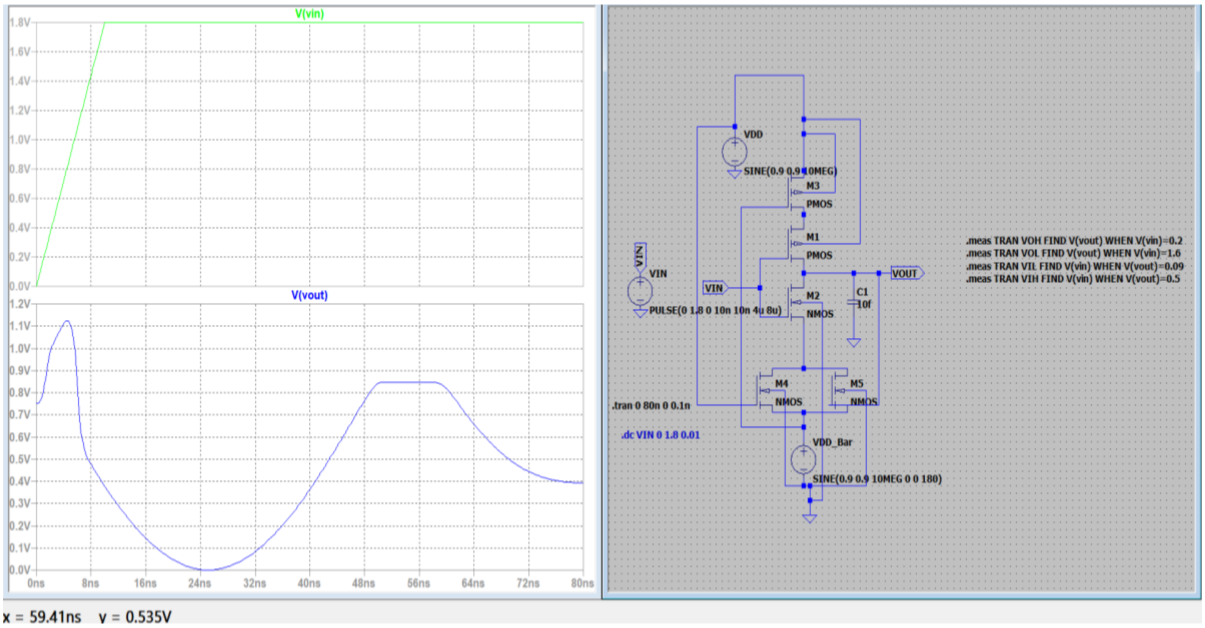


Figure 5 - Circuit and waveform of Noise Margin

### Noise Margin Value

SPICE Output Log: C:\Users\ASUS\OneDrive\Documents\LTspice\noise margin in energy recovery.log

```
Start Time: Sat Oct  4 15:29:45 2025
solver = Normal
Maximum thread count: 16
tnom = 27
temp = 27
method = trap
Instance "M2": Length shorter than recommended for a level 1 MOSFET.
Instance "M2": Width narrower than recommended for a level 1 MOSFET.
Instance "M4": Length shorter than recommended for a level 1 MOSFET.
Instance "M4": Width narrower than recommended for a level 1 MOSFET.
Instance "M5": Length shorter than recommended for a level 1 MOSFET.
Instance "M5": Width narrower than recommended for a level 1 MOSFET.
Instance "M1": Length shorter than recommended for a level 1 MOSFET.
Instance "M1": Width narrower than recommended for a level 1 MOSFET.
Instance "M3": Length shorter than recommended for a level 1 MOSFET.
Instance "M3": Width narrower than recommended for a level 1 MOSFET.
Direct Newton iteration for .op point succeeded.
Total elapsed time: 0.247 seconds.

Files loaded:
C:\Users\ASUS\OneDrive\Documents\LTspice\noise margin in energy recovery.net
C:\Users\ASUS\AppData\Local\LTspice\lib\cmp\standard.mos

voh: V(vout)=0.818626503423 at 1.11111114324e-09
vol: V(vout)=0.43151576954 at 8.88888885173e-09
vil: V(vin)=1.79999995232 at 3.22892823124e-08
vih: V(vin)=1.79999995232 at 6.91140668713e-08
```

Figure 6 - Value of Noise Margin

### 5)Switching Threshold Voltage through VTC Curve:

The switching threshold voltage is the input voltage at which the output of a digital inverter switches states, typically seen as the point on the Voltage Transfer Characteristic (VTC) curve where the output voltage equals the input voltage. This point marks the boundary between logical high and low output levels during the input voltage transition. It is critical for defining circuit stability and noise margins, as a balanced switching threshold is usually near half the supply voltage for optimal performance.

### Circuit Diagram

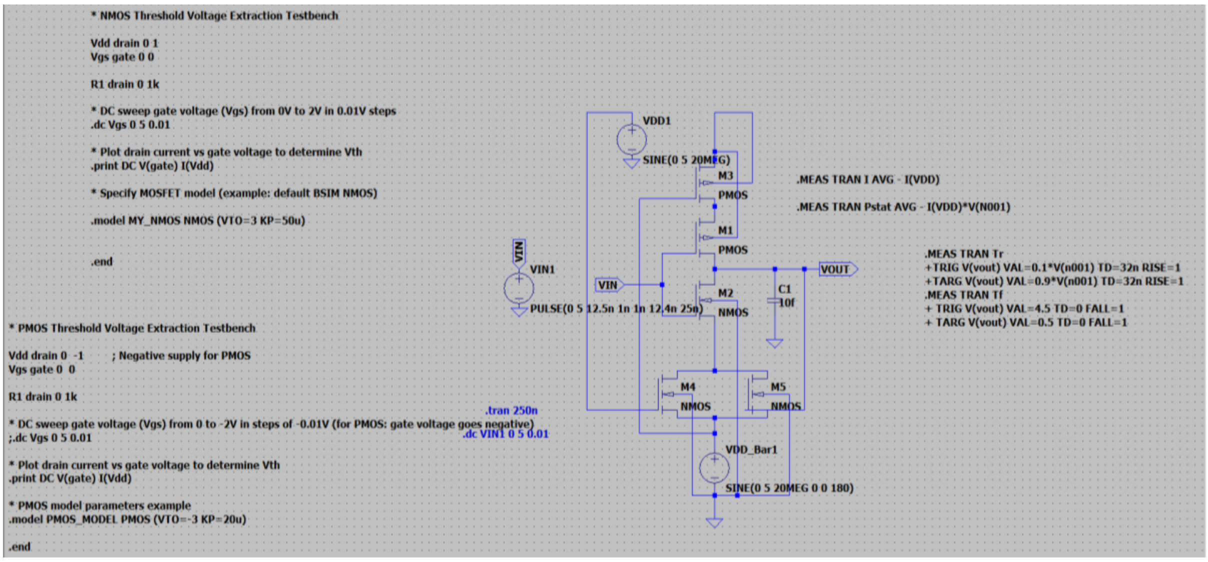


Figure 7 - Circuit Diagram of Switching Threshold Voltage through VTC curve

### Waveform


Figure 8 - Waveform of Switching Threshold Voltage

## 6)Static Power Consumption and Dynamic Power Consumption:

Static power consumption is the power used by a digital circuit when it is not switching, mainly due to small leakage currents flowing through transistors. Dynamic power consumption happens when the circuit switches states, charging and discharging capacitances, which uses energy. Simply put, static power drains energy even when idle, while dynamic power is used during active operations.

- **Static Power Consumption**= 3.9305959855×10^15 μW
- **Dynamic power Consumption**= −8.622286252×10^14 μW

**Circuit Diagram with Waveform**

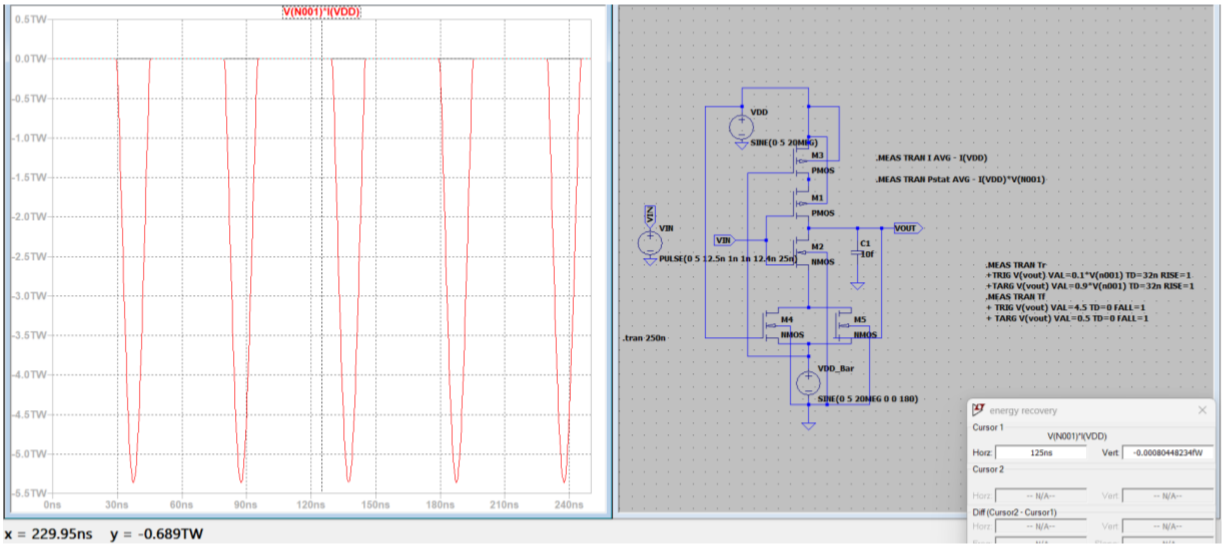


Figure 9 - Circuit and Waveform of power consumption of the Static and Dynamic

**Power**

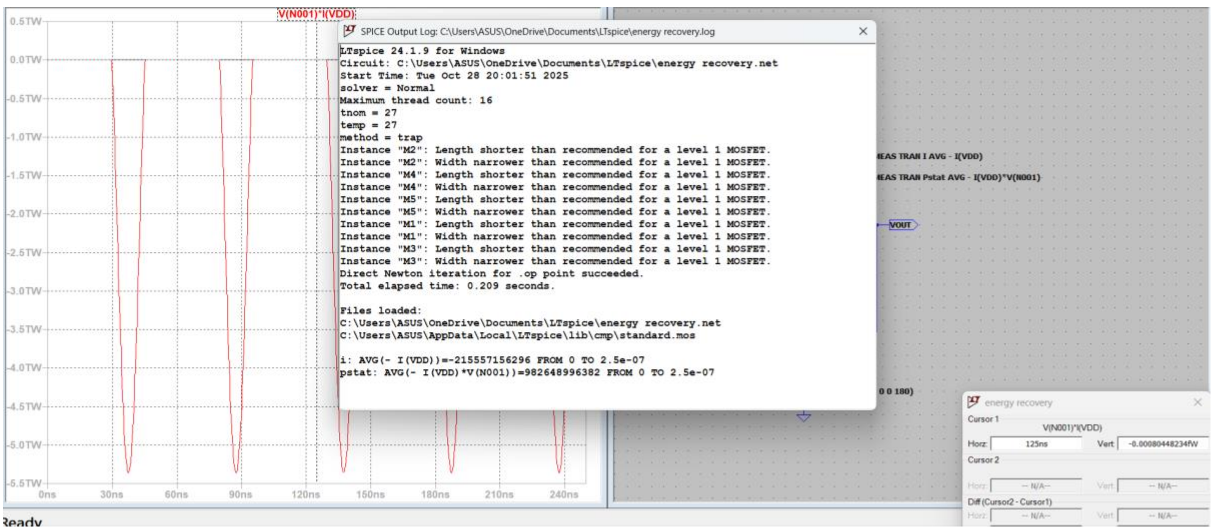


Figure 10 - Power value of Static and Dynamic

## 7)Power Reduction Technique:

Clock gating reduces power consumption by selectively turning off the clock signal to parts of a digital circuit that are not in use, preventing unnecessary switching activity. This cuts down dynamic power dissipation caused by charging and discharging of capacitive loads in idle blocks. By stopping the clock to inactive modules, clock gating significantly lowers overall power consumption without affecting circuit performance.

### Conventional Circuit Power :

- Static= 982648996382 from 0 to 2.5 e^(-7)
- Dynamic= -215557156296 from 0 to 2.5 e^(-7)
-

### Power Reduction Technique using Clock Gating

- Static= 982262928318 from 0 to 2.5 e^(-7)
- Dynamic= -215499917108 from 0 to 2.5 e^(-7)

### Circuit Diagram using Clock Gating

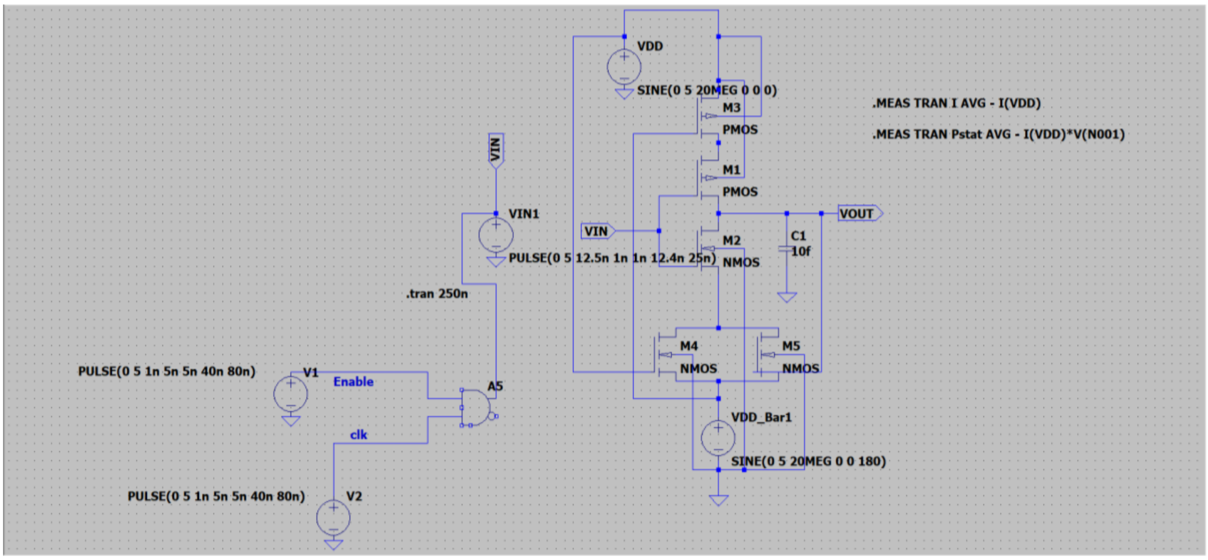


Figure 11 - Circuit diagram of Power reduction technique using clock gating

## Waveform

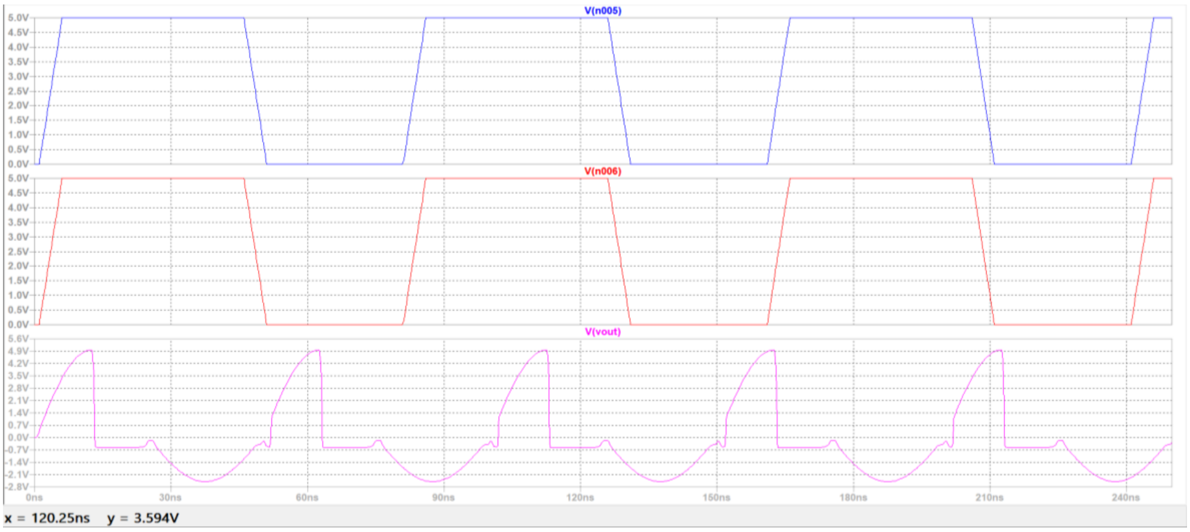


Figure 12 - Waveform of Power Reduction Technique

## Power

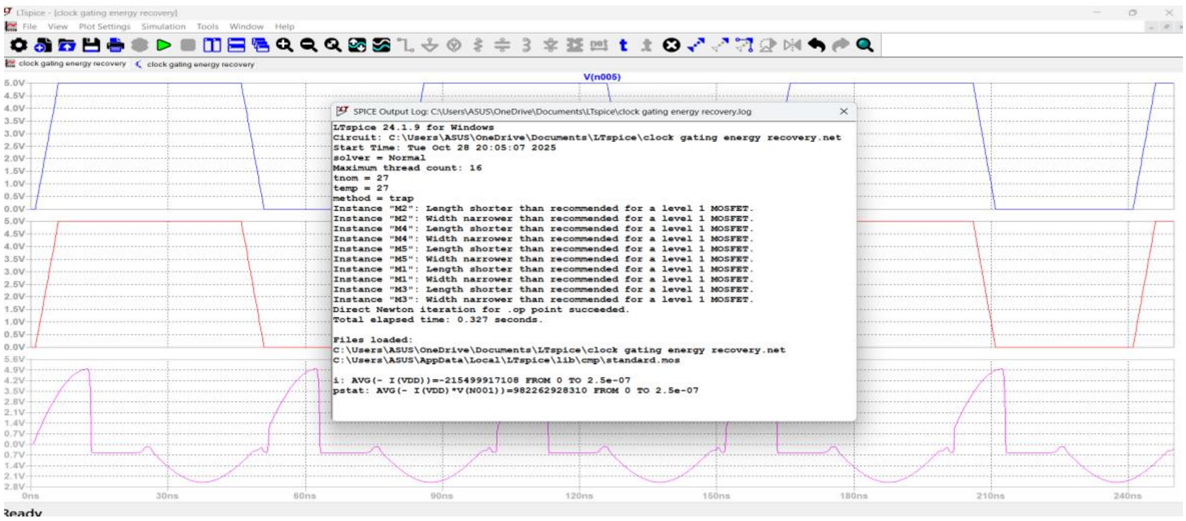


Figure 13 - Value of Power Reduction Technique

## 8)Level Convertor using Multi VDD design Concept (5 domains):

A Level Converter using Multi VDD design with 5 voltage domains divides a chip into regions operating at different supply voltages (voltage islands). These converters translate signal levels safely between low and high voltage domains to prevent static current flow and ensure correct logic interfacing. Multi-threshold transistors and carefully designed switching help reduce power consumption and delay at the interfaces of these domains.

### Circuit diagram

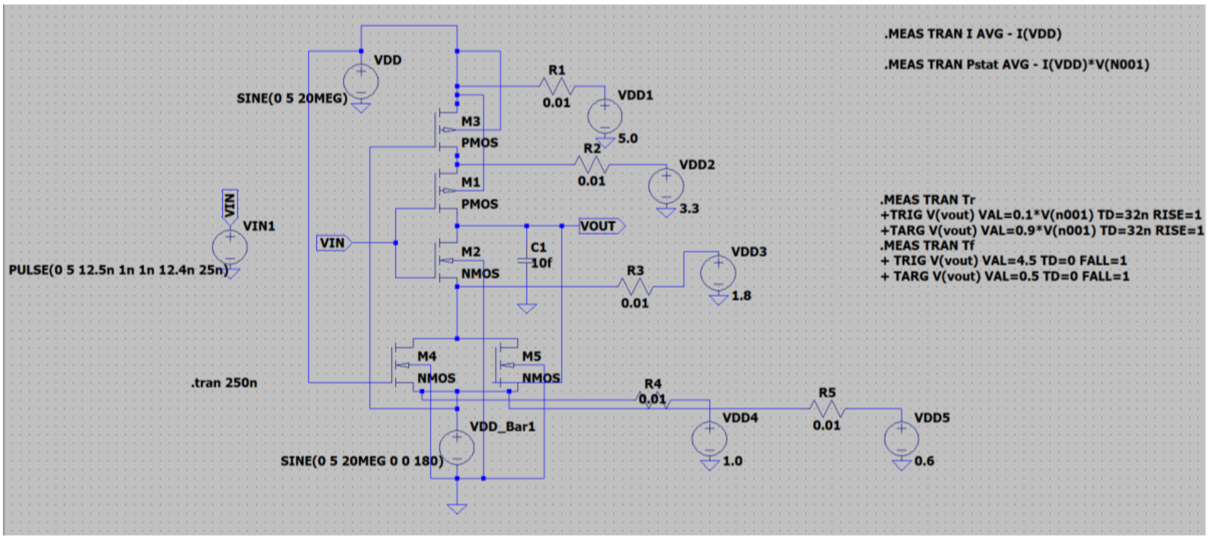


Figure 14 - Circuit diagram of Level Convertor using Multi Vdd

### Waveform

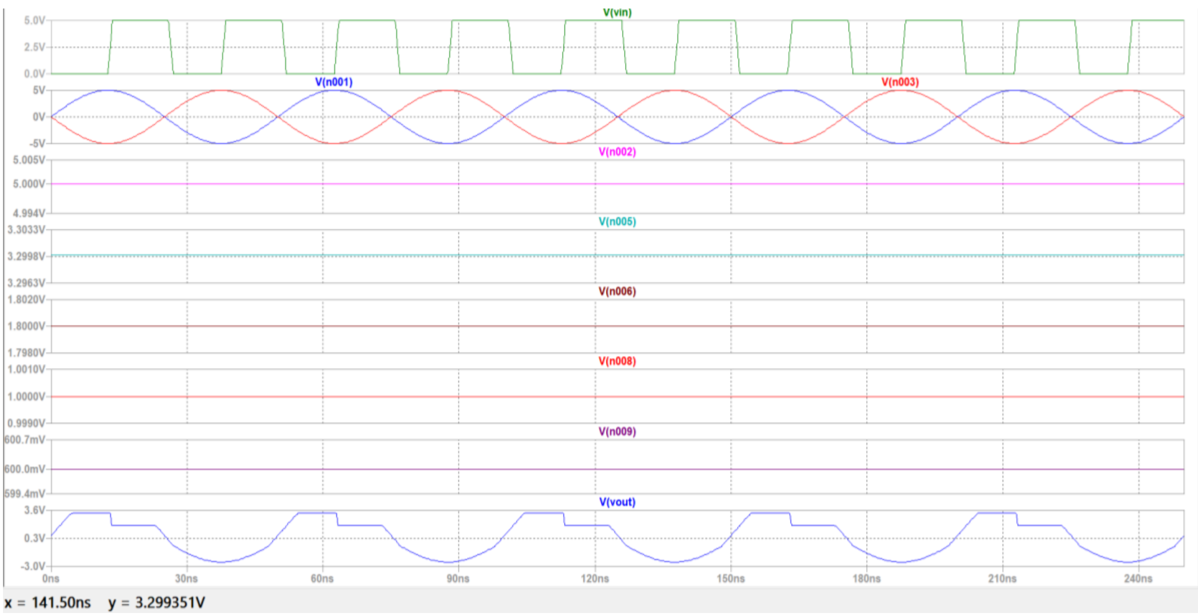

Figure 15 - Waveform of Level Convertor using multi Vdd

## 9)Case study: Application of energy recovery circuit in real time system:

Real-time systems can apply energy recovery circuits effectively to improve energy efficiency by reclaiming and reusing energy that would otherwise be lost. These circuits find practical use in various fields, including industrial automation, motor equipment, compressed air systems, and VLSI low power design.

**Application of Energy Recovery Circuits in Wearable Biomedical Devices**

Wearable biomedical devices, such as health monitors, pacemakers, and implantable sensors, require extremely low power consumption to ensure prolonged operation, reliability, and patient safety. The integration of energy recovery circuits based on improved diode-free adiabatic logic presents a transformative solution to meet these demands, enabling these devices to operate more efficiently and sustainably.

**The Need for Energy Efficiency in Wearable Healthcare**

Wearable health monitoring systems continuously collect vital physiological data—such as ECG, blood oxygen levels, and heart rate—and transmit this data for analysis. These systems are often battery-powered, and frequent battery replacements or recharges can hinder utility, increase maintenance costs, and cause discomfort or health risks to patients. Therefore, enhancing power efficiency is critical not only for extending battery life but also for ensuring the system's stability and accuracy over extended periods.

**Role of Energy Recovery Circuits**

Energy recovery circuits leverage adiabatic logic principles to reduce power dissipation during switching activities. Unlike traditional CMOS logic, which dissipates energy as heat because charges are irreversibly drained into the ground during switching, adiabatic logic circuits recycle part of this energy. The key mechanisms involve carefully timed, slowly varying power supplies and reversible charge flow, which enables the system to recover and reuse energy stored in load capacitances.

In wearable biomedical devices, such circuits enable sustained powering of crucial components such as sensors, amplifiers, and processors with minimal energy loss. The improved diode-free adiabatic logic circuitry ensures symmetric transistor sizing, which maintains signal integrity with high noise margins and controlled rise and fall times, essential for precise biomedical measurements.

**Benefits in Wearable Devices**

**1.Extended Battery Life:**

By recycling energy during each switching cycle, these circuits significantly reduce the overall power consumption. This directly translates to extended operational periods—potentially days or even weeks—without the need for battery replacement or recharging, which is vital for implantable devices such as pacemakers and long-term health monitors.

**2.Reduced Thermal Dissipation:**

Lower power dissipation means less heat generation, crucial in implantable or close-to-skin devices to prevent tissue damage and enhance patient comfort. The minimized heat also improves the reliability and lifespan of the device components.

**3.Enhanced Reliability and Stability:** Maintaining high noise margins and stable switching thresholds ensures accurate signal processing, para- critical in biomedical applications, where precise data acquisition can impact diagnosis and treatment outcomes.

**4.Compatibility with Multi-Vdd and Level Conversion Techniques:**

The use of multi-voltage domains (multi-Vdd) allows different parts of the system to operate at varying voltages, tailoring energy consumption to specific functional needs. Level converters enable communication between these domains without signal degradation, further optimizing energy use.

**Practical Implementation and Future Outlook**

Implementing energy recovery circuits in wearable biomedical devices involves optimizing transistor sizing, clocking schemes, and voltage domain management for reliable and efficient operation. The continuous evolution of adiabatic logic extends their applicability in ultra-low power health monitoring systems.

Future advancements may include integrating energy harvesting techniques—such as thermoelectric or RF energy harvesting—to supplement power supplies, creating self-sustainable biomedical devices. Additionally, research into miniaturized, integrated energy recovery modules promises greater adoption in compact, wearable health technology, enabling truly autonomous, long-term health monitoring solutions.

## Conclusion:

To conclude on the energy recovery circuit using improved diode-free adiabatic logic, the key findings highlight the circuit's efficiency in reducing power consumption through energy recovery techniques, involving optimized transistor sizing and symmetric configurations. The rise and fall times indicate the circuit's switching speed, while the noise margins (VIL and VOH) and switching threshold from the VTC curve ensure stable operation within designed voltage levels. Power analysis reveals significant static and dynamic power reductions compared to conventional circuits, especially when employing power-saving techniques such as adiabatic switching and multi-Vdd design to implement weak inversion regions across multiple voltage domains. The application of level converters demonstrates the feasibility of weak inversion operation in complex multi-Vdd systems, effectively extending the energy recovery benefits to broader applications.

**Reference**


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